

Qatar-1b

R-0641

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_230815 · created 2026-07-25T19:06:45+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460171.8264 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.189 +/- 0.055
Transit depth (Rp/Rs)^2	3.58 +/- 2.09 [%]
Orbital Inclination (inc)	80.14 +/- 2.93
Airmass coefficient 1 (a1)	0.371 +/- 0.093
Airmass coefficient 2 (a2)	0.79 +/- 0.36
Scatter in the residuals of the lightcurve fit is	27.04 %
Best Comparison Star	#1 - [372, 277]
Optimal Method	PSF photometry
Transit Duration (day)	0.035 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.189 ± 0.055 vs 0.1463 (deviation 0.78σ)
Duration fit vs published	0.035 d vs 0.0692 d (deviation 1.14σ)
Mid-time O-C	-6.5 min
Depth SNR	3.4
Residual scatter	27.04 %

Light curve

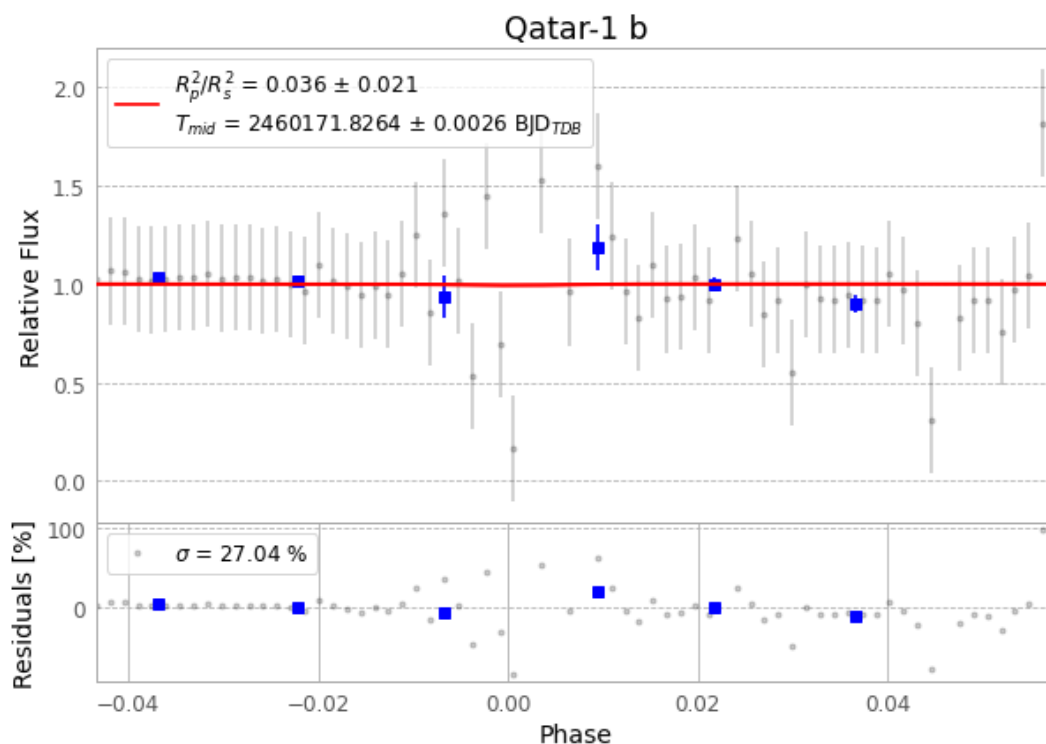


Figure 1 — FinalLightCurve_Qatar-1 b_2023-08-14.png (SHA-256 6396036828bab35d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [256, 198], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 26083eede017b6e9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

70 FITS frames (46 MB), Qatar-1230815061815.FITS → Qatar-1230815094515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230815.log (f0efb0b16352...), FinalLightCurve_Qatar-1 b_2023-08-14.png (6396036828ba...), FinalLightCurve_Qatar-1 b_2023-08-14.csv (fbc8703e3d78...)

Compute resources

Wall time 165.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 19:06Z.